# LIVESH L

**AI Developer | LLM Applications & Security | Machine Learning | Python & Backend Engineering**

Tamil Nadu, India | liveshjaidj@gmail.com | 9345406612 | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

Final-year B.Tech Artificial Intelligence & Data Science student (CGPA 8.53/10) focused on AI engineering, LLM applications, RAG systems, AI security, and production-oriented Python/FastAPI services. Built hybrid retrieval and context orchestration systems, open-source LLM security and reliability frameworks, and automated CI/CD pipelines. Experienced with PostgreSQL/pgvector, Docker, GitHub Actions, machine learning, and computer vision.

## TECHNICAL SKILLS

**AI/ML & LLMs:** Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, PyTorch, TensorFlow, scikit-learn, NLP, LLMs, RAG, CAG, MAG, hybrid retrieval, reranking, evaluation

**AI Security:** Prompt-injection defense, guardrails, PII/secret protection, policy-based authorization, threat modeling, multi-tenant isolation, RBAC

**Languages:** Python, Java, SQL, C#, TypeScript

**Backend & Frontend:** FastAPI, Flask, REST APIs, SSE, JWT, React, Node.js, Express.js

**Data & Cloud:** PostgreSQL/pgvector, MySQL, SQLite, Docker, Docker Compose, Kubernetes, AWS S3, Firebase, Render, GitHub Actions

**Tools & Practices:** Git, GitHub, Postman, Linux, pytest, Ruff, Agile/Scrum, OOP, DBMS, OS, Computer Networks, System Design

## EXPERIENCE

**AI Intern — Chargebee | Feb 2026 – Aug 2026**

- Built Leviathan, a sustainability and anti-greenwashing audit platform using a scikit-learn ensemble to predict carbon footprint and durability (reported $R^2$ = 0.88).
- Developed a Greenwashing Inspector using TF-IDF and Gemini reasoning to audit marketing claims against ISO 14021; integrated Open Food Facts and Google Shopping APIs.

**XR Developer — XARC | May 2026 – Aug 2026**

- Built the XARC Nexus Aggregator desktop client with React, TypeScript, Electron, and Unity Addressables for on-demand XR module loading.
- Configured AWS S3 for remote asset hosting/version management and implemented module management, download tracking, and update detection.

**Machine Learning Intern — Femtosoft Technologies | Jul 2025 – Aug 2025**

- Applied supervised and unsupervised ML across data collection, preprocessing, EDA, feature engineering, training, and evaluation.

**Power BI Intern — Cognifyz Technologies | Sep 2024 – Oct 2024**

- Built dashboards and reports; used Power Query, DAX measures, KPI analysis, filters, and drill-through reporting.

## SELECTED PROJECTS

**NexaRAG — Enterprise Document Intelligence & RAG Platform** GitHub

**Tech:** Python, FastAPI, PostgreSQL/pgvector, React, TypeScript, Docker, BM25, cross-encoder reranking, NeMo Guardrails.

- Built a multi-strategy RAG/CAG/MAG platform with hybrid retrieval, Reciprocal Rank Fusion, reranking, citation-grounded responses, SSE streaming, and tenant isolation.
- Implemented JWT/RBAC, prompt-injection defenses, secret/PII protection, audit logs, and document ingestion; reported 321 passing tests and ~24.5 ms average retrieval latency.

**MedicalTransformerLM — Medical LLM & RAG Workbench** GitHub

**Tech:** Python, PyTorch, FastAPI, Ollama, BM25, FAISS, Whisper, Unity/VR.

- Developed a local-first biomedical AI workbench combining a custom ~110M-parameter medical Transformer, OpenBioLLM, Llama 3.2, retrieval-grounded responses, voice interaction, and Unity VR integration.
- Added grounding checks, source metadata, medical acronym expansion, and fallback behavior for an educational/research prototype.

**LLMFirewall Core — Open-Source LLM Security Framework** GitHub

**Tech:** Python, FastAPI/ASGI, security engineering, policy enforcement, pytest.

- Built defense-in-depth controls for prompt injection, PII/secrets exposure, unsafe tool calls, RAG document poisoning, and policy-as-code enforcement.
- Implemented modular detection, risk scoring, redaction, audit logging, integrations, and automated security/regression validation.

**AI Reliability Engine (aireliability) — LLM Regression Testing** GitHub | PyPI

**Tech:** Python, Pydantic, pytest, Ruff, GitHub Actions, PyPI.

- Created deterministic AI-agent reliability tests for tool ordering, arguments, outputs, schemas, latency, cost, execution traces, and regression detection.
- Published the package to PyPI and automated CI/CD validation; resume project records 362 passing tests.

**AEGIS Framework — AI Reliability & Controlled Remediation** GitHub | PyPI

**Tech:** Python, FastAPI, Pydantic, Prometheus, OpenTelemetry, Loki, Kubernetes, Docker.

- Developed an open-source framework for incident investigation, evidence correlation, root-cause reasoning, and approval-gated remediation.
- Added fail-closed policies, bounded autonomy, dry-run actions, kill switches, recovery verification, and cryptographic audit trails.

**FastAPI CI/CD & Containerized Deployment Pipeline** GitHub

**Tech:** FastAPI, Python, Docker, GitHub Actions, Ruff, pytest, PostgreSQL.

- Built a production-style API pipeline with automated testing, linting, container builds, GHCR publication, and deployment workflows for cloud hosting.

**Stock Price Prediction** GitHub | Live Demo

**Tech:** Python, Pandas, NumPy, scikit-learn, TensorFlow/Keras, Matplotlib.

- Trained time-series forecasting models using historical market data and optimized a two-layer stacked LSTM; resume-reported results include under 4% MAPE across the top 100 blue-chip stocks.

**Skill Ascend — AI-Powered Career Mentorship** GitHub | Live Demo

**Tech:** React, TypeScript, D3.js, Node.js, Express.js, Gemini AI, Tailwind CSS.

- Built personalized career roadmaps, skill-gap-based learning curricula, XP/progress visualizations, and an AI mock-interview experience.

## EDUCATION

**B.Tech, Artificial Intelligence and Data Science** — Prathyusha Engineering College, Thiruvallur

Aug 2023 – Expected 2027 | CGPA: 8.53/10 | No active backlogs

## CERTIFICATIONS

OpenCV Bootcamp — OpenCV University (100%); Joy of Computing Using Python — NPTEL, IIT Madras (Elite 77%); Intro to Machine Learning — NPTEL; Cloud Computing — NPTEL; Databases and SQL for Data Science — IBM/Coursera; Power BI Micro Course — Learn More Pro.